

Projected Results: What Each Pending Experiment Would Buy

Planning memo for *Marginal Matching Does Not License Factorized Sampling*

How to read this memo. Numbers in **black** are measured results already in `main_v6.tex`. Numbers in **blue** are *projected best-case outcomes* for experiments that have not been run — they are reasoned forecasts anchored on the measured numbers, not data. This memo exists to answer one question: *for each pending experiment, what would we most likely get, what would it cost, and what would it buy the paper?* Nothing here should be copied into the paper; the paper reports only measured values.

1. Summary and recommended order

Compute estimates use the observed per-run cost on this machine (a 300-epoch F-CS-WAE run took 4.5–13 h wall-clock depending on contention from other tenants; call it ~ 8 h; diagnostics are seconds). The key planning fact: **the four cheapest experiments require no training at all**, take roughly one working day in total, and close four of the reviewer’s P0/P1 gaps.

Experiment	Compute	Buys	Risk if it goes badly
<i>Tier 1 — no training required (~ 6–8 h total)</i>			
Post-hoc sampling controls	1–2 h	Kills the τ -shrinkage confound	Low
JointMMD calibration	1–2 h	Upgrades term-4 claim to established	Medium
Support-aware swap	2–3 h	Kills the “off-support” objection	Medium
Intra-class diversity metric	1 h	Turns Exp. 4 assumption into measurement	Low
<i>Tier 2 — light training (~ 8 h)</i>			
Two external classifiers	4–8 h	Closes the #1 evaluation gap	High
<i>Tier 3 — heavy training</i>			
Multi-seed core (3 seeds)	~ 128 GPU-h ≈ 3 d	Removes single-seed caveat	Medium
<i>Tier 4–5 — new code + training</i>			
Invariance baselines (GRL/HSIC/VFAE)	~ 1 d dev + 2 d	Highest scientific payoff	High
Joint remedy for term 4	~ 1 d dev + 1 d	A method contribution	High

Recommendation. Tier 1 + Tier 2 (~ 1 – 1.5 days) is strictly the best value: it closes four P0/P1 gaps and independently validates the paper’s most-quoted numbers. Tier 3 (multi-seed) is the expensive long pole but removes the caveat attached to almost every result. Tier 4 is the only item that could raise the paper’s ceiling rather than just its floor — see §6.

2. Tier 1a — Post-hoc sampling controls

Question. Is the post-hoc prior’s gain from *class conditioning*, or merely from *variance shrinkage* ($\tau=0.25$ samples near the posterior mean, where the decoder behaves well)?

Measured anchor (MNIST, $\delta=0$, self-accuracy): global $\mathcal{N}(0, I) = 0.150$; class-mean = 1.000; class-diag $\tau=0.25 = 1.000$; $\tau=0.50 = 1.000$; empirical bank = 1.000.

Control (new)	Isolates	Projected self-ACC
Global fitted Gaussian, $\tau=0.25$	shrinkage, <i>no</i> conditioning	0.12–0.20
Shuffled-label class-conditional	<i>wrong</i> conditioning	0.09–0.14
Per-class mean + global covariance	conditioning, no shrinkage	0.97–1.00
Class-conditional, $\tau=1.0$	conditioning, no shrinkage	0.82–0.93

Projected conclusion. Shrinkage without conditioning stays near the 0.150 baseline; conditioning without shrinkage recovers nearly all of the gain. This resolves the confound in the paper’s favour. The reason confidence is high: the measured class-mean strategy is conditioning with *near-zero* variance and already reaches 1.000, while a globally-fitted prior carries no class information at all and therefore cannot select the right class by construction.

3. Tier 1b — JointMMD calibration (term 4)

Measured anchor. MNIST 0.00372 ($\delta=0$) $\rightarrow$ 0.00432 ($\delta=1$); CIFAR-10 0.00359 $\rightarrow$ 0.00435. Currently reported as *suggestive only*: one biased estimate per checkpoint, no null, no error bars.

- Permutation null (double within-class permutation): 0.0003–0.0012
- Observed vs. null: significantly above, $p < 0.01 \Rightarrow$ term 4 is genuinely non-zero
- $\delta=0$ vs. $\delta=1$ difference (+0.0006): not significant ($\pm\text{CI} \approx 0.0008$)
- Conditional HSIC cross-check: same direction, confirms

Projected conclusion. Term 4 is non-zero and is not reduced by per-class style MMD — upgrading the paper’s central theoretical prediction from “consistent with” to “established”. **Risk:** if the null comes out close to the observed value, the result is inconclusive and the estimator itself needs work (per-block normalization, bandwidth ladder).

4. Tier 1c/1d — Support-aware swap; intra-class diversity

Support-aware swap. Answers the standing objection that cross-class (z_c, z_s) pairs may simply be off the training joint’s support.

Pair type	Support	Projected content-following
Same image	in-support	0.99–1.00
Same class, cross image	in-support	0.93–0.98
Cross class	off-support	0.001 (measured)
Interpolation α	—	monotone, crossover $\alpha \approx 0.3$ –0.45
kNN dist. to train joint	—	cross-class 1.5–2.5 $\times$ same-class

Projected conclusion. Same-class swaps — which are firmly in-support — preserve class identity, while cross-class swaps do not. That dissociation is what the paper needs: it shows style-dominance is not merely an artifact of leaving the support. **Risk:** if cross-class pairs turn out drastically farther from the joint, part of the objection survives and the claim must stay hedged.

Intra-class diversity. DINOv2/CLIP within-class mean pairwise feature distance, projected: MNIST 0.28–0.38 < Fashion-MNIST 0.42–0.52 < CIFAR-10 0.60–0.72 — the assumed ordering confirmed with clear separation, turning Experiment 4’s assumption into a measurement. (Even so, three datasets remain too few to call it a correlation.)

5. Tier 2 — External classifiers (the #1 gap)

Today, generation self-accuracy *and* the latent-swap rates are scored by the model’s own auxiliary classifier, and the swap re-encodes through the model’s own encoder — a shared bias could inflate both.

Quantity	Measured (internal)	Projected (external)
External classifier real-test acc	—	$\sim 99.4\%$ MNIST; $\sim 94\%$ CIFAR-10
MNIST naive self-ACC	0.150	0.13–0.18
MNIST class-cond. $\tau=0.25$	1.000	0.95–0.99
MNIST swap style-following	99.2%	94–99%
CIFAR-10 class-cond. $\tau=0.25$	0.508	0.35–0.45
Two-classifier agreement	—	93–97% MNIST; 80–88% CIFAR-10

Projected conclusion. MNIST conclusions survive essentially intact; CIFAR-10 numbers soften materially. This is the honest risk worth flagging to the reviewer in advance: an external classifier is stricter than the model’s own on blurry FID-83 CIFAR-10 samples, so Experiment 5’s headline numbers (68.1% for $\tau=0.25$, 95.9% for the empirical bank) would likely need restating *downward*. The qualitative claims (leakage exists; the recipe transfers poorly to CIFAR-10) would not change — if anything a downward revision *strengthens* the negative transfer result.

6. Tier 4 — Invariance baselines: the one experiment that could raise the ceiling

Measured anchor. Per-class style MMD at $\delta=1$ (MNIST): LP 100% $\rightarrow$ 42.6%, Δ_{inter} 6.21 $\rightarrow$ 1.22, naive self-ACC 0.150 $\rightarrow$ 0.735.

Method	Projected LP	Projected Δ_{inter}	Projected naive self-ACC
Per-class style MMD (measured)	42.6%	1.22	0.735
VFAE-style conditional MMD	42–50%	1.1–1.4	0.70–0.78
HSIC penalty (calibrated)	30–45%	0.9–1.3	0.72–0.82
Gradient-reversal adversary	15–28%	0.6–1.1	0.70–0.85

Why this is the highest-value experiment. The projected GRL row is the interesting one: GRL is expected to drive LP down toward chance (15–28%, far below per-class MMD’s 42.6%) while naive generation self-accuracy still fails to reach 1.0. If that holds, it is a direct *empirical* confirmation of the paper’s central theoretical claim — that driving term 2 to zero is *not sufficient* for valid factorized sampling, because term 4 survives (Corollary 1). That would promote Theorem 1 from an organizing device to a validated prediction, which is exactly the gap the reviewer identified.

The real risk, stated plainly. If GRL drives LP to chance *and* naive self-accuracy to ≈ 1.0 , then term 2 alone *was* the whole story on this model, and the paper’s emphasis on term 4 weakens considerably. This experiment can cut either way — which is precisely why it is worth running, and why it should be run before the term-4 remedy of Tier 5 is built on top of it.

7. Tier 3 — Multi-seed, and the two outliers it would resolve

Measured anchor (single seed per point). Clustering ACC is non-monotonic in δ , with exactly one large outlier per dataset:

δ	0	0.03	0.1	0.3	1	3
MNIST ACC	0.992	0.994	0.990	0.994	0.859	0.993
CIFAR-10 ACC	0.813	0.811	0.810	0.715	0.800	0.799

- MNIST $\delta=1$ over 3 seeds: 0.95 ± 0.06 — the 0.859 absorbed as seed variance
- CIFAR-10 $\delta=0.3$ over 3 seeds: 0.79 ± 0.04 — likewise
- Δ_{inter} std across seeds: ± 0.05 –0.20 (smooth metric, low variance)
- LP std across seeds: ± 1 –3 points
- Swap content-following at $\delta=1$: 58–64% (measured single-seed: 60.9%)

Projected conclusion. Both outliers dissolve into seed variance, yielding the clean claim the paper currently cannot make: *leakage falls smoothly with δ while clustering cost stays small and roughly flat.* **Risk:** if the outliers reproduce, there is a real δ -dependent training instability that needs its own explanation — also a publishable observation, but a different paper section.

8. Tier 5 — Joint remedy for term 4

Only worth building if Tier 4 confirms term 2 is insufficient.

- $\mathcal{L}_{\text{joint}}$ or cHSIC($z_c, z_s \mid y$): JointMMD 0.0037 $\rightarrow$ 0.0012–0.0020
- Swap content-following: 60.9% $\rightarrow$ 72–85%
- MNIST naive self-ACC: 0.735 $\rightarrow$ 0.85–0.95
- CIFAR-10 naive self-ACC: 0.366 $\rightarrow$ 0.50–0.65

Projected conclusion. A remedy targeting term 4 beats per-class MMD on naive factorized sampling, on both datasets — converting the paper from an audit into an audit + method. **Risk:** joint distribution matching is hard to optimize; a plausible outcome is a marginal improvement that does not justify the added complexity.

9. Bottom line for the reviewer's scorecard

The reviewer scored the current draft 5/10, blocked on three gaps: non-independent evaluation, an uncalibrated term-4 measurement, and confounded/overstated repair claims.

- **Tier 1 + Tier 2** ($\sim 1\text{--}1.5$ days): closes the term-4 calibration gap, the τ -shrinkage confound, the off-support objection, and the evaluation-independence gap. **Projected:** 5 $\rightarrow$ 6/10.
- + **Tier 3** (~ 3 days): removes the single-seed caveat attached to nearly every table. **Projected:** 6 $\rightarrow$ 6.5/10 (borderline).
- + **Tier 4** (~ 3 days incl. dev): supplies the missing empirical validation of Theorem 1 and the direct invariance comparison. **Projected:** $\rightarrow$ 7/10 (weak accept).

Total to a defensible weak-accept: $\approx 7\text{--}8$ days of wall-clock, of which only ~ 1.5 days sits on the critical path before the paper's claims stop being hedged.